

Check Device Overheat Tool Guide

Revision History

Date	Version	Author	Note
2026-08-11	1.0	Hau Pham	Initial version

Prerequisites

- **Python 3.10 installed** (via the `py` launcher, so `py -3.10` resolves) — pinned, not "3.10 or later": the pinned `PyQt5/PyQtWebEngine/matplotlib` wheel versions in `requirements.txt` aren't published for newer Python versions (e.g. 3.14). Other Python versions can coexist on the same machine; 3.10 just needs to be one of them.
- AWS SSO access (profile `fw-btcy-sso`). Check session validity with:

```
aws sts get-caller-identity --profile fw-btcy-sso
```

A printed account/ARN means the session is valid.

- If expired, the tool will show `Error when retrieving token from sso: Token has expired` when calling SSH/S3. Fix: run

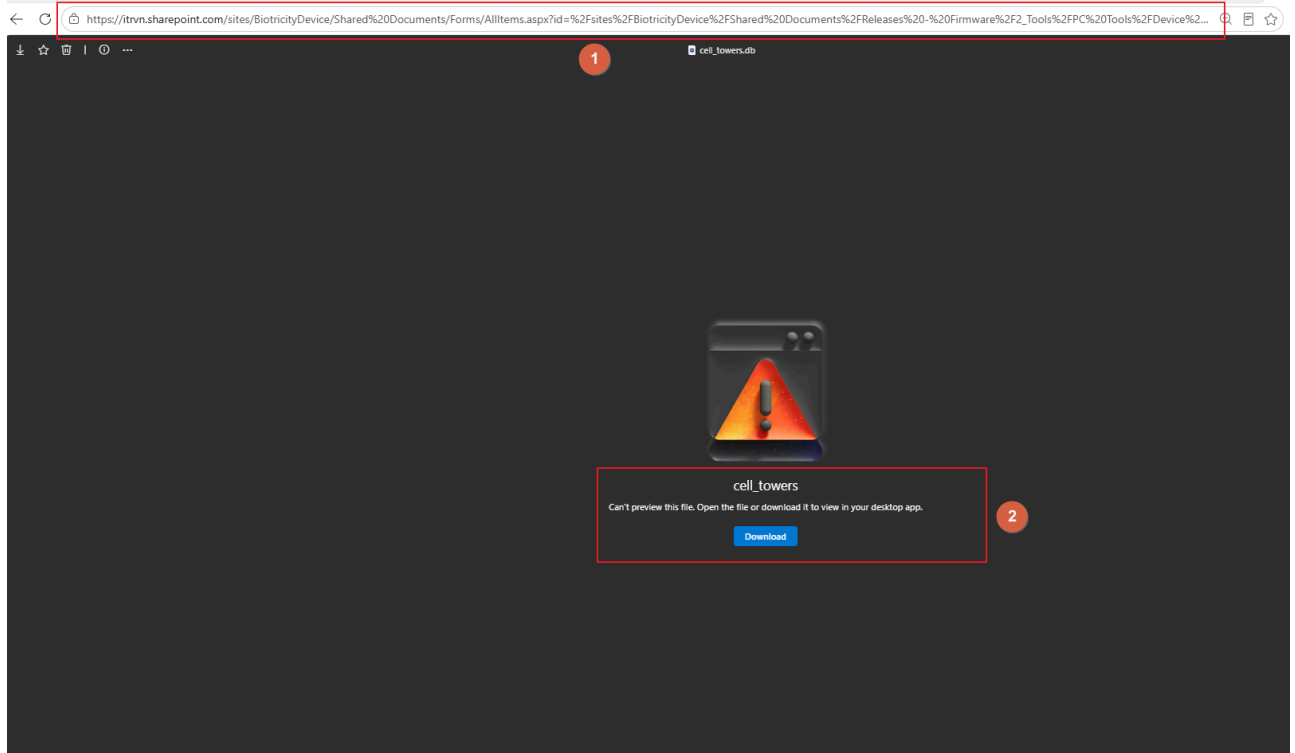
```
aws sso login --profile fw-btcy-sso
```

Open the link, enter the code printed, sign in via the browser, then retry.

- `setup.bat` already run once in `check-overheat-tool/` (creates `.venv` + installs dependencies):

```
setup.bat
```

If `database/cell_towers.db` (the local cell-tower lookup DB, ~595MB — needed for the Map tab) isn't there yet, `setup.bat` opens its SharePoint download page and **fails** — download can't be scripted (needs an interactive Microsoft login):



① Note the page (can't preview a `.db` file), ② click **Download**. Move the downloaded file to `check-overheat-tool\database\cell_towers.db` (create the `database` folder if needed), then re-run `setup.bat` to verify.

- Launch the tool:

```
run.bat
```

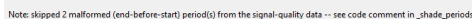
Tab Reference

General overview of every control on the tool's 4 tabs. For an actual usage flow, see the [Example 1](#) / [Example 2](#) walkthroughs below.

"Add Device" tab

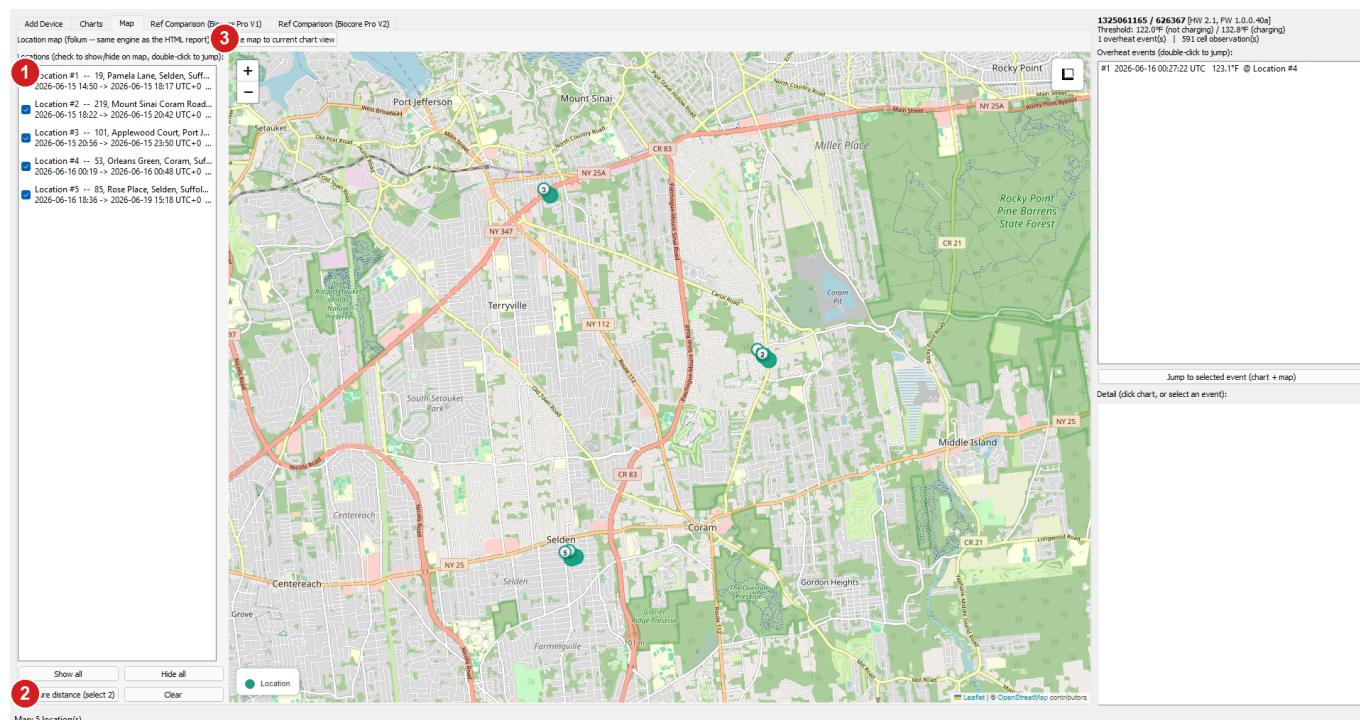
Reference Comparison: loaded from cache (nothing changed since last computed) | excluded 313 sample(s) within 3 min of a charging<->discharging flip from the min/max/avg stats (transitional, non-issue devices only)

- "Charts" tab



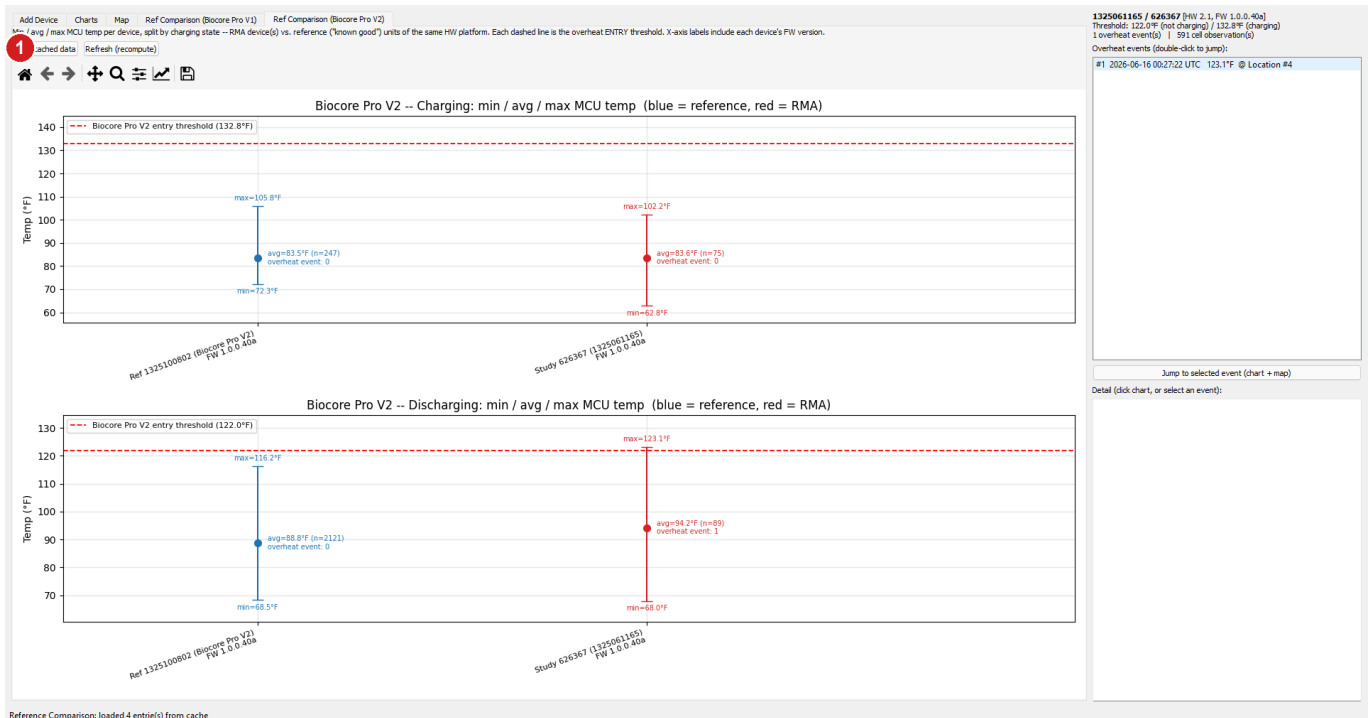
1. **Full range** / Hours before/after — zoom controls (the double-click-to- event zoom window size).
2. **Not-charging entry/exit, Charging entry/exit** — show/hide the threshold-crossing markers on the Temperature chart.
3. **Cellular (device log)** — off by default (slower on long studies); fills in brief cellular wakeups the server-reported mode misses.
4. **Back-cover (est.)** — an estimated skin-contact temperature line, not a real per-device measurement.
5. **Overheat events** — double-click to jump; **Jump to selected event**.
6. **Detail** — click the chart, or select an event.

"Map" tab



1. **Locations** — check/uncheck to show/hide, double-click to jump, **Show all** / **Hide all**.
2. **Measure distance (select 2)** — straight-line, not driving distance — / **Clear**.
3. **Update map to current chart view** — limits the map to whatever the Charts tab is zoomed to.

"Ref Comparison" tabs



One tab per HW platform (V1 / V2 — different thresholds, never mixed). Each device's min/avg/max MCU temp vs. a reference ("known good") device, split by charging state; dashed line = overheat entry threshold.

1. **Load cached data** (instant, no recompute) / **Refresh (recompute)** (slow with many devices).

Example 1: "HOT DEVICE REPORTED — Device ID 1325061165 CareMedPrimary and Urgent Care PC (Heart Smart Group)"

HOT DEVICE REPORTED - Device ID 1325061165 CareMedPrimary and Urgent Care PC (Heart Smart Group) Tung Nguyen Dai Tran Hau Pham

Summarize this email

Sent: Tuesday, 16 June 2026 12:07:33
 To: Al Young <ayoung@biotricity.com>, Pareeha Siddiqui <psiddiqui@biotricity.com>
 CC: Janis Hardill <jhardill@biotricity.com>, Spencer LaDow <sladow@biotricity.com>
 Subject: HOT DEVICE REPORTED - Device ID 1325061165 CareMedPrimary and Urgent Care PC (Heart Smart Group)

Hi,

In covering support, had this call below from QCSS - and below is my conversation with the patient:

Called the pt. and she advised that when she put the device on yesterday afternoon - 2pm by 8:00 on the device itself was warm and where it made contact with her skin she said it was very hot to the touch. She states that there was a red mark where this was - she is going to email me a picture of the device. (will forward a copy of that when received) She did confirm that the red mark was from the device and not the electrodes. The device itself was hot against the skin but the top part of the device felt warm but not as hot as the back. She is going to return the device to the clinic and get a new device. She did state that the device cooled down once she powered the device down.

Missy Neumann
 Customer Success Manager

missyneumann@biotricity.com

440 668 5387



Email: alyzabailey@sw@gmail.com

From: Kerrill J. Schmidt <kjschmidt@qcossinc.com>
 Sent: Tuesday, June 16, 2026 8:52 AM
 To: Biotricity Customer Service <cs@biotricity.com>
 Subject: QCSS/Biotricity Escalation

Caller's Name: Ms Alyza Bailey
 Patient's Name: Ms Alyza Bailey
 Clinic: Care Med in Seldon, NY
 Device Type: Biocore
 Phone: 631 707 3897

Reason For Escalation: Patient stated the Biocore device is very hot. Advised to take device off. Please have Biotricity call Patient Ms. Alyza Bailey at 631 707 3897.

Kerrill Schmidt
 Client Services Manager
 QCSS, Inc.
 21925 W. Field Parkway, Suite 210
 Deer Park, IL 60010
kjschmidt@qcossinc.com
www.qcossinc.com



Device ID **1325061165**, email sent 2026-06-16 → the incident = "yesterday" = 2026-06-15, around 2pm-8pm.
No RMA yet (patient hasn't returned the device).

Step 1 — "Add Device" tab: enter Device ID + approximate date

Since there's no RMA yet, use the **"or by date"** field instead of "Look up RMAs":

- 1. **Device ID** field: enter 1325061165.
- 2. **"or by date"** field: pick 2026-06-15 (the date inferred from "yesterday" in the email).
- 3. Click **"Search studies near this date"**.

Add DeviceChartsMapRef Comparison (Biocore Pro V1)Ref Comparison (Biocore Pro V2)

Add a study by Study ID, by Device ID to look up its RMAs first, or by Device ID + an approximate date if there's no RMA yet (e.g. a "hot device" report before the patient has returned it).

Input folder:W:\report\TOOL_CHECK_OVERHEAT_DEVICE\check-overheat-tool\input_log_demoBrowse...

Study ID:friendly ID or full hashAdd this Study ID

Device ID:1325061165Look up RMAs

or by date:2026-06-15Search studies near this date

Day threshold:- 45 days / + 45 days around the Date of Event ("Use this RMA" below) or the manual date above

RMA ID	Model	RMA Status	Canceled	Rejected	Requested At	Reason for RMA	Failure Analysis	RMA Resolution	IA Billing Resoluti	Date of Event	Patient Impact	Received	Received At	RMA Notes
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Use this RMA (search studies near its date)

Study candidates near the date above (nearest first) -- pick one, then Download:

Friendly ID	Start (UTC)	Stop (UTC)	Duration
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Download selected study

Log:2026-08-11 17:52:14.665 [INFO] Clearing study view -- no studies left in the picker2026-08-11 17:52:14.720 [INFO] Main window initialized -- 0 study(ies) available2026-08-11 17:52:14.720 [INFO] Reference Comparison redraw (force=False)Open log folder

Studies already added:

Remove selectedAdd study folder...Refresh signal/speed/locationLoad

Reference Comparisons: loaded from cache (nothing changed since last computed) | excluded 243 sample(s) within 3 min of a charging<->-discharging flip from the min/max/avg stats (transitional, non-issue devices only)

Step 2 — "Study candidates" table: pick the right study

The tool calls issue 494, lists every study for the device within ±45 days of the date entered, **ranked nearest-first** — it does NOT auto-download, allowing manual confirmation of the right one first.

Study 626367 (2026-06-15 18:04 → 2026-06-16 01:11 UTC) is ranked first — it lines up closely with "2pm" (18:04 UTC – 4h = 2:04 PM local). Select this row, then click **"Download selected study"**.

Add DeviceChartsMapRef Comparison (Biocore Pro V1)Ref Comparison (Biocore Pro V2)

Add a study by Study ID, by Device ID to look up its RMAs first, or by Device ID + an approximate date if there's no RMA yet (e.g. a "hot device" report before the patient has returned it).

Input folder:W:\report\TOOL_CHECK_OVERHEAT_DEVICE\check-overheat-tool\input_log_demoBrowse...

Study ID:friendly ID or full hashAdd this Study ID

Device ID:1325061165Look up RMAs

or by date:2026-06-15Search studies near this date

Day threshold:- 45 days / + 45 days around the Date of Event ("Use this RMA" below) or the manual date above

RMA ID	Model	RMA Status	Canceled	Rejected	Requested At	Reason for RMA	Failure Analysis	RMA Resolution	IA Billing Resoluti	Date of Event	Patient Impact	Received	Received At	RMA Notes
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Use this RMA (search studies near its date)

Study candidates near the date above (nearest first) -- pick one, then Download:

Friendly ID	Start (UTC)	Stop (UTC)	Duration
1 57	2026-06-15 18:04	2026-06-16 01:11	7h 7m
2 62693	2026-06-08 13:35	2026-06-14 03:25	5d 13h 49m
3 630008	2026-06-22 21:08	2026-06-25 21:08	3d
4 633318	2026-06-29 18:21	2026-07-06 18:21	7d

Download selected study

Log:2026-08-11 17:52:14.665 [INFO] Clearing study view -- no studies left in the picker2026-08-11 17:52:14.720 [INFO] Main window initialized -- 0 study(ies) available2026-08-11 17:52:14.720 [INFO] Reference Comparison redraw (force=False)2026-08-11 17:52:16.870 [INFO] -- Manual date search: device 1325061165, around 2026-06-15 (no RMA) ...2026-08-11 17:52:16.888 [INFO] Fetching studies for device 1325061165 via issue 494 (-45d/+45d around 2026-06-15)...2026-08-11 17:52:45.911 [INFO] Found 5 candidate study(ies), nearest first.Open log folder

Studies already added:

Remove selectedAdd study folder...Refresh signal/speed/locationLoad

Reference Comparisons: loaded from cache (nothing changed since last computed) | excluded 243 sample(s) within 3 min of a charging<->-discharging flip from the min/max/avg stats (transitional, non-issue devices only)

Step 3 — After downloading, run "Refresh signal/speed/location"

Once downloaded, the study shows up under **"Studies already added"**. At this point the Temperature chart already has data, but **RSRP/RSRQ/Upload Speed/Map are still empty** — the just-downloaded logs are only enough for the temperature chart; RSRP/speed/location need a separate step.

Select that row in the list, then click **"Refresh signal/speed/location"** (can take a few minutes — it runs issue 495 to fetch server logs, then resolves RSRP/speed/cell-location for this specific device).

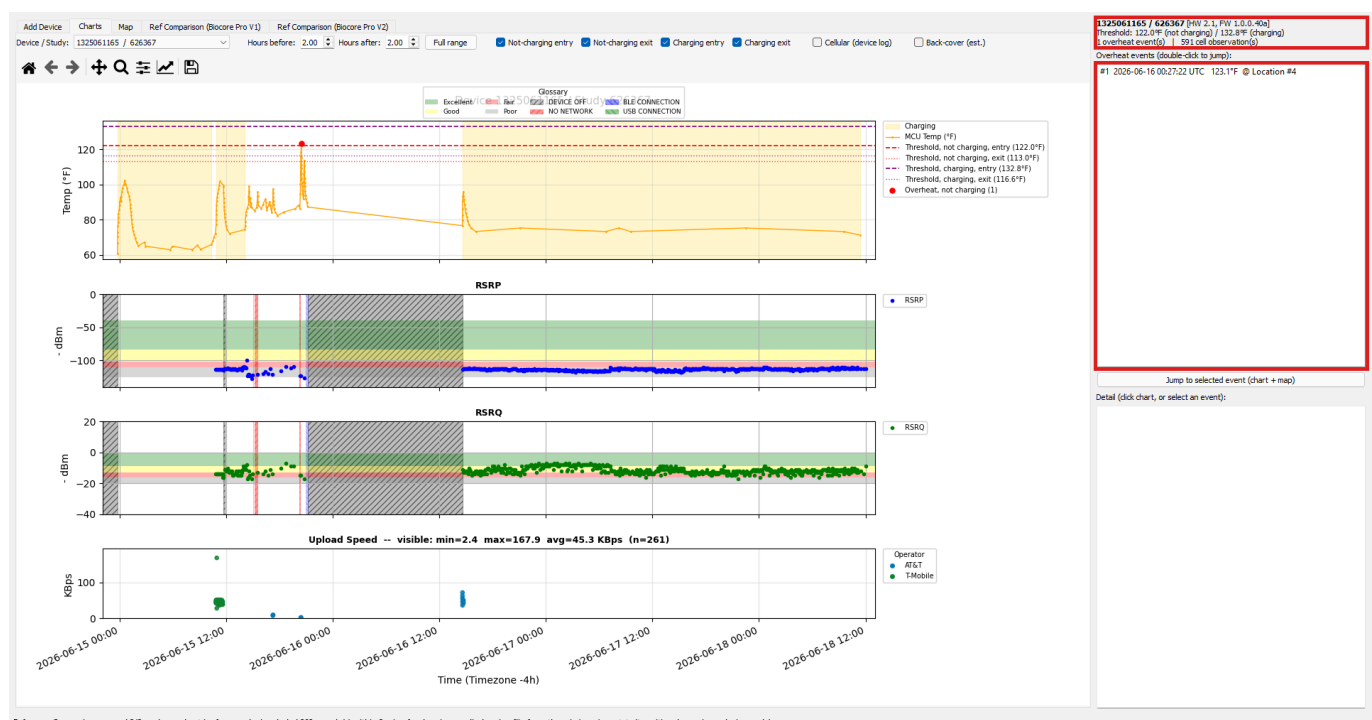
1 361165 / study 626367 -- RMA ()

2 Refresh signal/speed/location

Reference Comparison: reused 2/3 unchanged entries from cache | excluded 260 sample(s) within 3 min of a charging<->discharging flip from the min/max/avg stats (transitional, non-issue devices only)

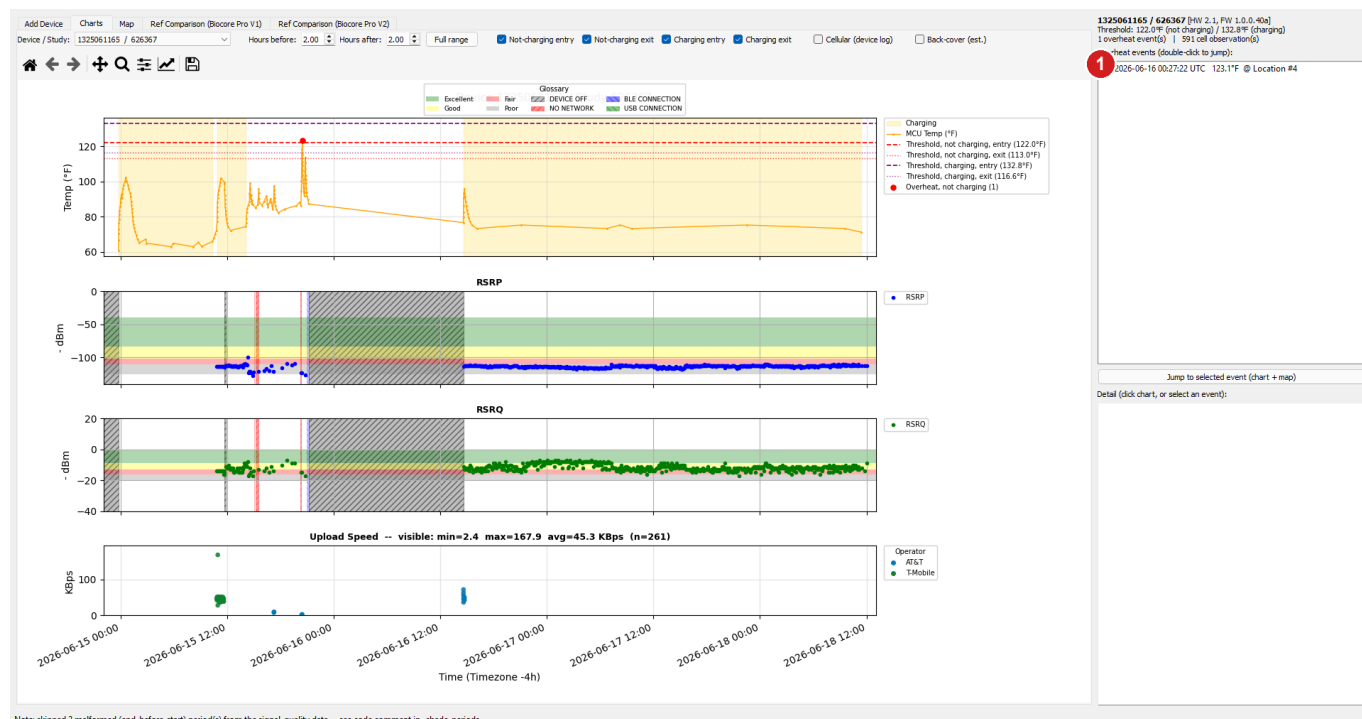
Step 4 — "Charts" tab: investigate the overheat event

First, read the FW version and threshold straight off the info panel (top right):



- **HW 2.1, FW 1.0.0.40a**
- **Threshold: 122.0°F (not charging)** — a reduced threshold (vs. 131°F on FW 1.0.0.38c — that comparison number comes from general firmware knowledge, not something the tool computes).
- **1 overheat event: 2026-06-16 00:27:22 UTC, 123.1°F, @ Location #4.**

① **Double-click the event** in the list to zoom the chart right into it:



Zoomed in, the story becomes clear:



- Right before the overheating, RSRP/RSRQ are sitting in the **Poor/Fair** band (bottom bands) and upload speed is low (a handful of points under 10 KBps) — network quality was poor, so the modem had to work harder/longer to get data out, which is what drove the temperature up to the threshold.
- Right after the event, the device goes into a long gray **"DEVICE OFF"** band — the user powered it off.

Click **"Full range"** (top toolbar) to zoom back out — in the earlier overview screenshot above, notice that **after** the DEVICE OFF gap ends, RSRP/RSRQ jump back into the **Good/Excellent** band and upload speed picks back up — network quality recovered once the device came back on.

② **Switch to the Map tab** to see *where* that recovery happened — see Step 5.

Step 5 — "Map" tab: confirm the location change, measure the distance

The study has 5 locations total, but only **#4** (where the overheat happened) and **#5** (where the device resumed) are relevant to this incident. Declutter the map first:

①② ③ **Uncheck Location #1, #2, #3** (leaves only #4/#5 visible on the map — these 3 are just earlier stops the patient passed through that day, unrelated to the overheat):

The screenshot displays the 'Map' tab of a mobile device monitoring application. The top navigation bar includes 'Add Device', 'Charts', 'Map', and two 'Ref Comparison' links for 'Biocore Pro V12'. Below the navigation bar, there are instructions: 'Location map (Forum - same engine as the HTML report):' and 'Update map to current chart view'. A list of locations is shown on the left, with locations 1, 2, and 3 unchecked, and locations 4 and 5 checked. The map area is currently empty. The details panel on the right shows information for Location #4, including coordinates, RSRP, and a list of events.

Location map (Forum - same engine as the HTML report): Update map to current chart view

Locations (check to show/hide on map, double-click to jump):

- Location #1 -- 15, Pamela Lane, Selden, Suff...
2026-06-15 14:50 -> 2026-06-15 18:17 UTC+0 ...
- Location #2 -- 219, Mount Sinai Coram Road...
2026-06-15 18:22 -> 2026-06-15 20:42 UTC+0 ...
- Location #3 -- 101, Applewood Court, Port J...
2026-06-15 20:56 -> 2026-06-15 23:50 UTC+0 ...
- Location #4 -- 53, Orleans Green, Coram, Suf...
2026-06-16 00:19 -> 2026-06-16 00:48 UTC+0 ...
- Location #5 -- 85, Rose Place, Selden, Suffol...
2026-06-16 18:36 -> 2026-06-16 19:18 UTC+0 ...

Show all Hide all

Measure distance (select 2) Clear

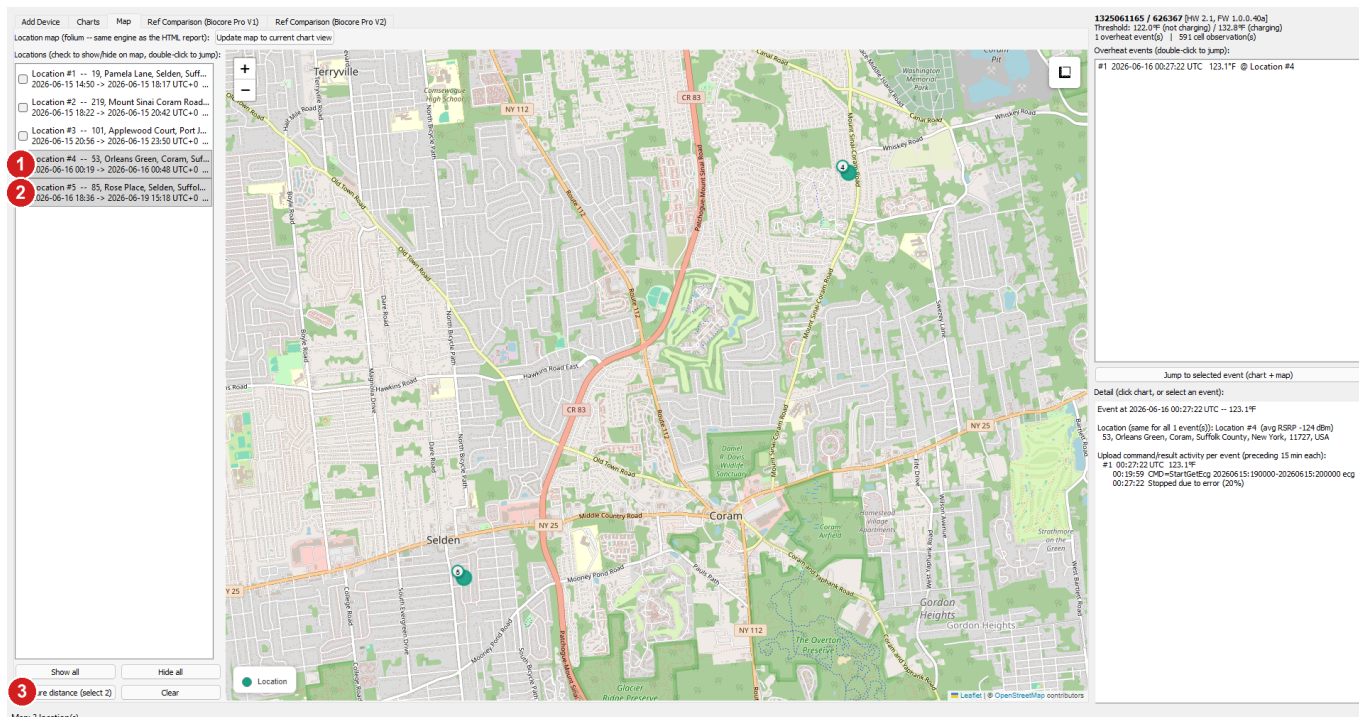
Jumped to 2026-06-16 00:27:22 UTC

1325061165 / 626367 [HW 2.1, FW 1.0.0.40a]
Threshold: 122.0°F (not charging) / 132.8°F (charging)
1 overheated event(s) | 391 cell observation(s)
Overheat events (double-click to jump):
#1 2026-06-16 00:27:22 UTC 123.1°F @ Location #4

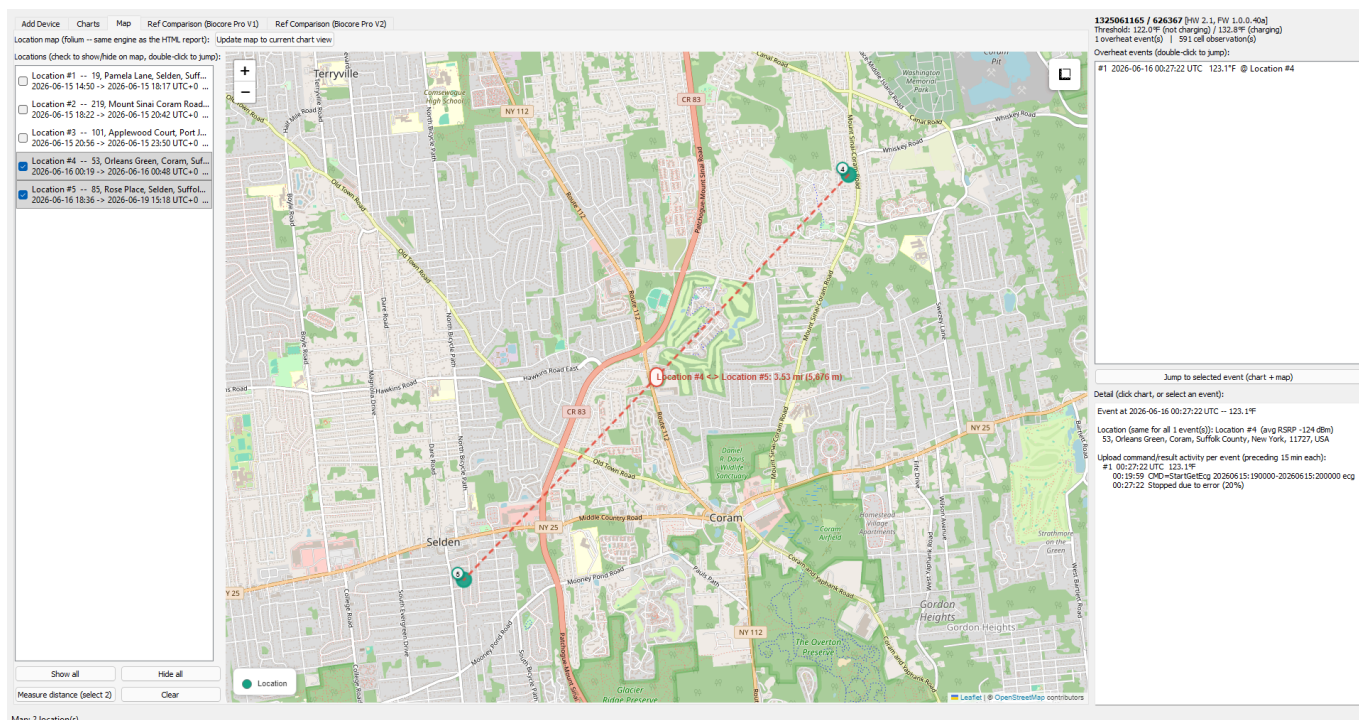
Jump to selected event (chart + map)

Detail (click chart, or select an event):
Event at 2026-06-16 00:27:22 UTC -- 123.1°F
Location (same for all 1 event(s)): Location #4 (avg RSRP -124 dBm)
53, Orleans Green, Coram, Suffolk County, New York, 11722, USA
Upload command/result activity per event (preceding 15 min each):
#1 00:27:22 UTC 123.1°F
00:19:59 CMD->StartGetEqg 20260616:190000-20260616:200000 eqg
00:27:22 Stopped due to error (20%)

① **Click Location #4**, ② **Ctrl/Shift-click Location #5** to select both (selecting is a separate action from the checkboxes above — checkboxes control what's *drawn* on the map, selection picks *which 2* to measure between), then ③ **click "Measure distance (select 2)":**



Result — a clean map with just the 2 relevant locations and the measured distance:



Location #4 <-> Location #5: 3.53 mi (5,676 m) — straight-line (great-circle) distance, which can come out slightly lower than the real driving distance (~3.67 mi, as written in the reply below) since real roads aren't a straight line.

This confirms the full picture: the device was at Location #4 when it overheated (poor signal there), the user powered it off, and by the time it powered back on it had physically moved ~3.5 miles away to Location #5 (where signal was back to normal) — **conclusion: signal quality was poor specifically at Location #4**, not a general device defect.

Putting it together → drafting the reply

Every figure below came directly from the tool:

We investigated the operational logs for Device ID 1325061165 during the timeframe when the patient reported the heating issue.

During this study, the device was running **FW version 1.0.0.40a** with a reduced MCU overheat threshold of **122°F** (previously 131°F in FW version 1.0.0.38c).

Our analysis indicates that the reported heating event occurred while the device was operating at **Location #4**. During this period, network quality and upload speed were poor, requiring the modem to operate at a higher workload for a longer duration to complete data transmission. This increased modem activity caused the device temperature to reach the configured MCU overheat threshold.

The user later powered off the device. On the following day, around 2:00 PM on June 16, the device resumed operation at **Location #5**, approximately **3.67 miles** from Location #4, where network quality and upload speed had returned to normal.

Example 2: "Required VN Assistance for BCP RMA Returns"

Required VN Assistance for BCP RMA Returns Hau Pham Summarize this email

Hi Al,

This is to acknowledge receipt of your email. The team will check and provide investigation results ASAP.

Thang

From: Al Young <ayoung@biotricity.com>
Sent: Tuesday, August 4, 2026 7:28 AM
To: Thang Nguyen (sys dev1) <sysdev1@biotricity.com>
Cc: Spencer LaDow <sladow@biotricity.com>; Amjad Talpur <atalpur@biotricity.com>
Subject: Re: Required VN Assistance for BCP RMA Returns

Sorry, Thang. I forgot to mention in my previous email that all three devices had received the firmware update controlling the temperature limit.

From: Al Young <ayoung@biotricity.com>
Sent: Monday, August 3, 2026 5:24 PM
To: Thang Nguyen (sys dev1) <sysdev1@biotricity.com>
Cc: Spencer LaDow <sladow@biotricity.com>; Amjad Talpur <atalpur@biotricity.com>
Subject: Required VN Assistance for BCP RMA Returns

Hello Thang:

We need VN support to investigate the following device RMAs. All three devices below were involved in customer complaints alleging that the patient was either burnt from the device or had experienced excessive heating. Two of the devices must be investigated to close our CAPA for the heating issues, and the remaining device must be investigated to close a complaint for a patient claiming to be burnt by the device.

RMA: 14474
Device ID: 1324050352
Issue Reported: Excessive heating claim

RMA ID: 14768
Device ID: 1325061716
Issue Reported: Same as above.

RMA ID: 15008
Device ID: 1325062089
Issue Reported: Patient claimed to be burnt by the device and provided pictorials.

To save time in having to send these devices, can the VN Team please review the data logs for the last study before the RMA was issued to verify any **overheating** anomalies?

One important note. Device ID 1325062089 (RMA 15008) was used on a second patient after the previous patient had claimed to be burnt by the device. Please review the log file for the previous patient connected to this device.

I will await your findings. Thanks.

Three devices, **all with known RMA IDs already** — unlike Example 1, the **"Look up RMAs"** flow can be used (Device ID → pick the RMA row → search studies near its Date of Event) instead of the manual "or by date" field:

RMA ID	Device ID	Issue Reported
14474	1324050352	Excessive heating claim
14768	1325061716	Same as above
15008	1325062089	Patient claimed to be burnt by the device, provided pictures

Device 1324050352 (RMA 14474) — Biocore Pro V1

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① Enter the Device ID, ② click "Look up RMAs":

Add DeviceChartsMapRef Comparison (Bicore Pro V1)Ref Comparison (Bicore Pro V2)

Add a study by Study ID, by Device ID to look up its RMAs first, or by Device ID + an approximate date if there's no RMA yet (e.g. a "hot device" report before the patient has returned it).

Input folder:W:\report\TOOL_CHECK_OVERHEAT_DEVICE\check-overheat-tool\input_log_demoBrowse...

Study ID:friendly ID or full hashAdd this Study ID

Device ID:130352Look up RMAs

or by date:2026-08-11Search studies near this date

Day threshold:- 45 days / + 45 days around the Date of Event ("Use this RMA" below) or the manual date above

RMA ID	Model	RMA Status	Cancelled	Rejected	Requested At	Reason for RMA	Failure Analysis	RMA Resolution	IA Billing Resoluti	Date of Event	Patient Impact	Received	Received At	RMA Notes
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Use this RMA (search studies near its date)

Study candidates near the date above (nearest first) -- pick one, then Download:

Friendly ID	Start (UTC)	Stop (UTC)	Duration
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Download selected study

Log:Open log folder

2026-08-11 18:01:51.253 [INFO] Clearing study view -- no studies left in the picker
2026-08-11 18:01:51.305 [INFO] Main window initialized -- 0 study(es) available
2026-08-11 18:01:51.306 [INFO] Reference Comparison redraw (force=False)

Studies already added:

Remove selectedAdd study folder...Refresh signal/speed/locationLoad

Reference Comparisons: loaded from cache (nothing changed since last computed) | excluded 243 sample(s) within 3 min of a charging<->discharging flip from the min/max/avg stats (transitional, non-issue devices only)

① Select the matching RMA row, ② click "Use this RMA":

Add DeviceChartsMapRef Comparison (Bicore Pro V1)Ref Comparison (Bicore Pro V2)

Add a study by Study ID, by Device ID to look up its RMAs first, or by Device ID + an approximate date if there's no RMA yet (e.g. a "hot device" report before the patient has returned it).

Input folder:W:\report\TOOL_CHECK_OVERHEAT_DEVICE\check-overheat-tool\input_log_demoBrowse...

Study ID:friendly ID or full hashAdd this Study ID

Device ID:1324050352Look up RMAs

or by date:2026-08-11Search studies near this date

Day threshold:- 45 days / + 45 days around the Date of Event ("Use this RMA" below) or the manual date above

RMA ID	Model	RMA Status	Cancelled	Rejected	Requested At	Reason for RMA	Failure Analysis	RMA Resolution	RMA Billing Resolution	Date of Event	Patient Impact	Received	Received At
1	Biotres Pro	Done	No	No	2026-06-16 17:54:14	DEVICE ERRORS DESCRIBE	Other: device is with QA, do not charge for RMA	NON REPAIRABLE	OTHER DESCRIBE: device is with QA, do not charge for RMA	2026-06-16	NONE	Yes	2026-07-20 21:49:31
2	11840	Biotres Pro	Done	No	2025-11-11 15:28:30	RETURN TO STOCK	NO ISSUE FOUND	Other: No issue found.	OTHER DESCRIBE: No issues found	2025-11-11	NONE	Yes	2026-01-07 00:12:38
3	10154	Biotres Pro	Done	No	2025-05-28 21:57:57	CLOSED ACCOUNT	Other: wrong charger	Other: No issue found; CHARGER REPLACED	OTHER DESCRIBE: No repairs needed	2025-05-28	NONE	Yes	2025-06-11 23:05:05
4	8353	Biotres Pro	Done	No	2024-10-18 13:36:59	RETURN TO STOCK	NO ISSUE FOUND	Other: No issue found.	OTHER DESCRIBE: No repairs needed	2024-10-18	NONE	Yes	2024-11-30 22:41:13

Use this RMA (search studies near its date)

Study candidates near the date above (nearest first) -- pick one, then Download:

Friendly ID	Start (UTC)	Stop (UTC)	Duration
-------------	-------------	------------	----------

Download selected study

Log:Open log folder

2026-08-11 18:01:51.253 [INFO] Clearing study view -- no studies left in the picker
2026-08-11 18:01:51.305 [INFO] Main window initialized -- 0 study(es) available
2026-08-11 18:01:51.306 [INFO] Reference Comparison redraw (force=False)
2026-08-11 18:01:53.195 [INFO] Looking up RMAs for device: 1324050352 ...
2026-08-11 18:01:53.445 [INFO] Found 4 RMA(s) for device: 1324050352.

Studies already added:

Remove selectedAdd study folder...Refresh signal/speed/locationLoad

Reference Comparisons: loaded from cache (nothing changed since last computed) | excluded 243 sample(s) within 3 min of a charging<->discharging flip from the min/max/avg stats (transitional, non-issue devices only)

This lists study candidates near the RMA's Date of Event (2026-06-16) — same ranked-table flow as Example 1. ① Pick the right one, ② Download:

Add DeviceChartsMapRef Comparison (Biocore Pro V1)Ref Comparison (Biocore Pro V2)

Add a study by Study ID, by Device ID to look up its RMA first, or by Device ID + an approximate date if there's no RMA yet (e.g. a "hot device" report before the patient has returned it).

Input folder:W:\report\TOOLS_CHECK_OVERHEAT_DEVICE\check-overheat-tool\input_log_demoBrowse...

Study ID:friendly ID or full hashAdd this Study ID

Device ID:1324050352Look up RMAs

or by date:2026-08-11Search studies near this date

Day threshold:- 45 days / + 45 daysaround the Date of Event ("Use this RMA" below) or the manual date above

RMA ID	Model	RMA Status	Canceled	Rejected	Requested At	Reason for RMA	Failure Analysis	RMA Resolution	RMA Billing Resolution	Date of Event	Patient Impact	Received	Received At
114474	Biocores Pro	Done	No	No	2026-06-16 17:54:14	DEVICE ERRORS DESCRIBE	Other: device is with QA, do not charge for RMA	NON REPAIRABLE	OTHER DESCRIBE: device is with QA, do not charge for RMA	2026-06-16	NONE	Yes	2026-07-20 21:49:31 AI, Spencer, Amjad e
211840	Biocores Pro	Done	No	No	2025-11-11 15:28:30	RETURN TO STOCK	NO ISSUE FOUND	Other: No issue found.	OTHER DESCRIBE: No issues found	2025-11-11	NONE	Yes	2026-01-07 00:12:38 Clinic notes...
310154	Biocores Pro	Done	No	No	2025-05-28 21:57:57	CLOSED ACCOUNT	Other: wrong charger	Other: No issue found; CHARGER REPLACED	OTHER DESCRIBE: No repairs needed	2025-05-28	NONE	Yes	2025-06-11 23:05:05
48353	Biocores Pro	Done	No	No	2024-10-18 13:36:59	RETURN TO STOCK	NO ISSUE FOUND	Other: No issue found.	OTHER DESCRIBE: No repairs needed	2024-10-18	NONE	Yes	2024-11-30 22:41:13

Use this RMA (search studies near its date)

Study candidates near the date above (nearest first) -- pick one, then Download:

Friendly ID	Start (UTC)	Stop (UTC)	Duration
133	2026-06-12 18:14	2026-06-15 18:14	3d
2623503	2026-06-09 15:20	2026-06-12 12:54	2d 21h 34m
3622300	2026-06-05 20:06	2026-06-07 23:19	2d 3h 13m
4620105	2026-06-02 16:16	2026-06-05 12:15	2d 19h 58m

Download selected study

Log:Open log folder

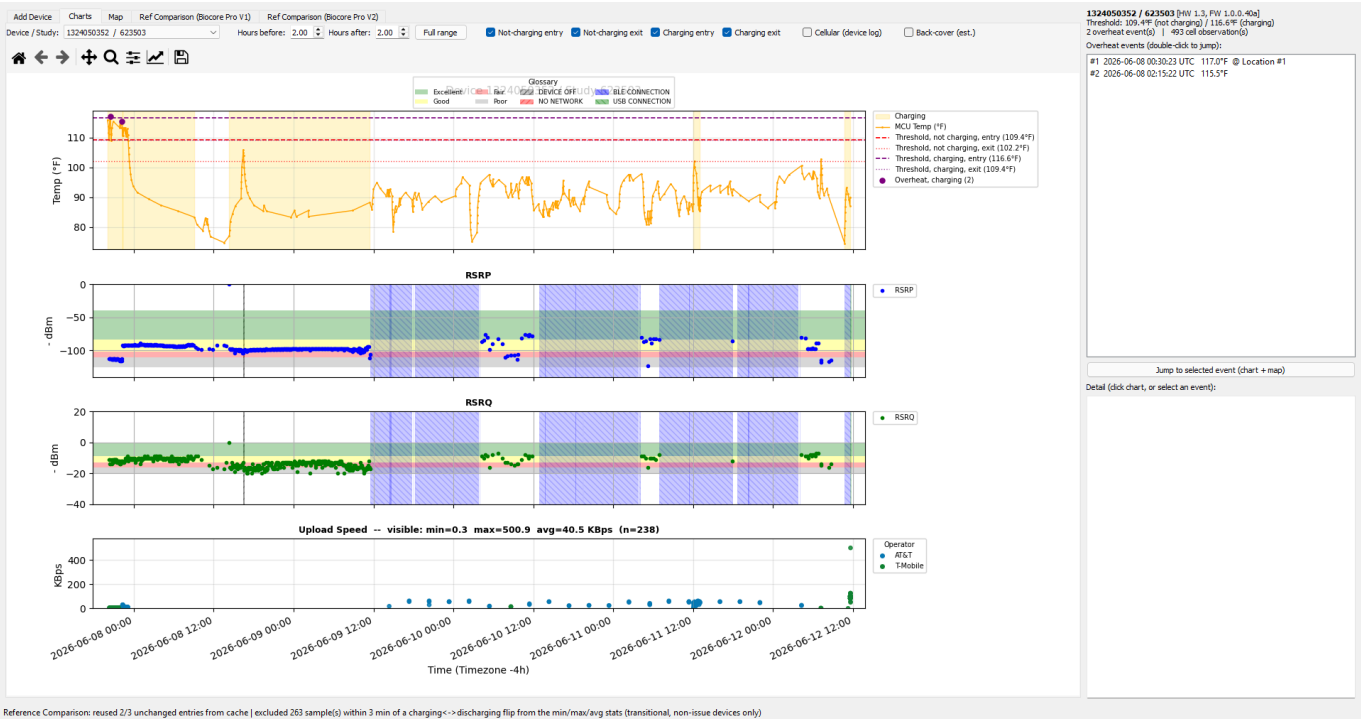
2026-08-11 18:01:51.253 [INFO] Clearing study view -- no studies left in the picker
2026-08-11 18:01:51.305 [INFO] Main window initialized -- 0 study(es) available
2026-08-11 18:01:51.306 [INFO] Reference Comparison (redraw force=false)
2026-08-11 18:01:53.195 [INFO] Looking up RMAs for device 1324050352 ...
2026-08-11 18:01:53.445 [INFO] Found 4 RMA(s) for device 1324050352
2026-08-11 18:01:54.084 [INFO] -- RMA 14474 / device 1324050352 (Date of Event 2026-06-16) --
2026-08-11 18:01:54.105 [INFO] Fetching studies for device 1324050352 via issue #94 (-45d/+45d around 2026-06-16) ...
2026-08-11 18:02:23.220 [INFO] Found 12 candidate study(es), nearest first.

Studies already added:

Remove selectedAdd study folder...Refresh signal/speed/locationLoad

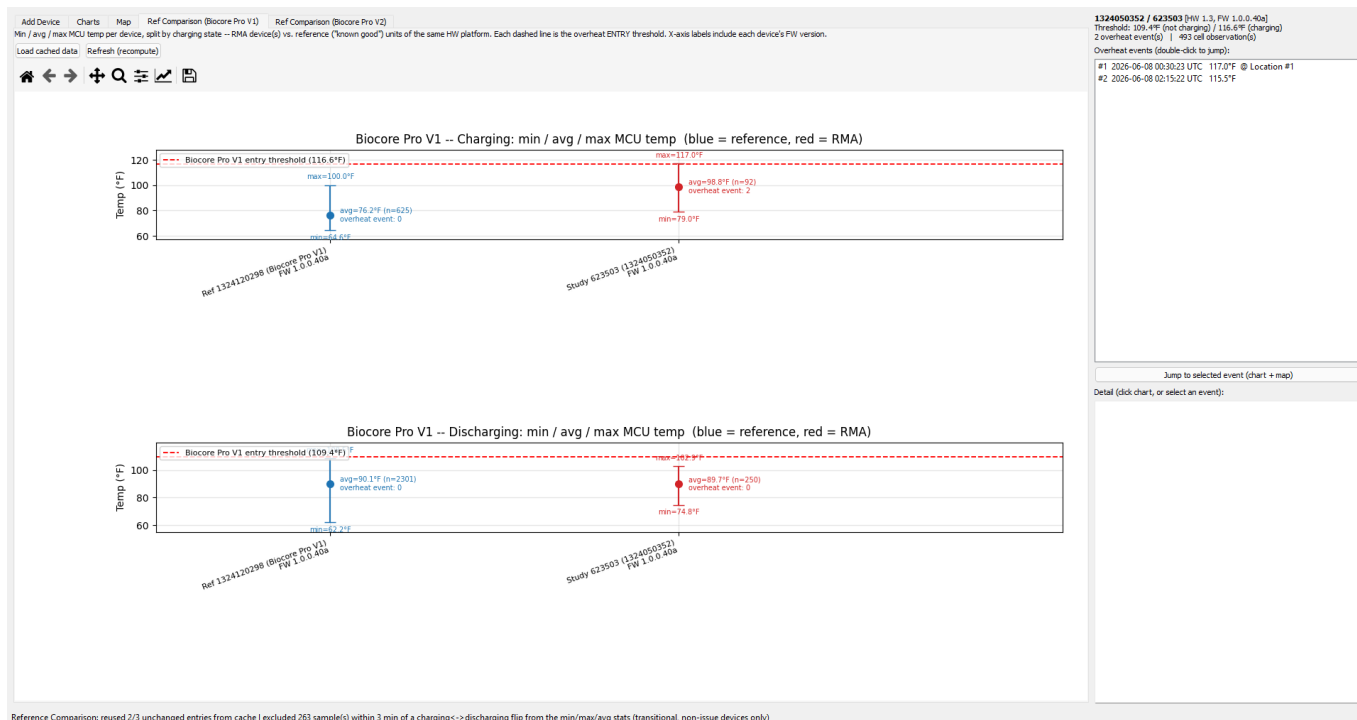
Reference Comparison: loaded from cache (nothing changed since last computed) | excluded 243 sample(s) within 3 min of a charging<->discharging flip from the min/max/avg stats (transitional, non-issue devices only)

Charts tab for 623503:



- HW 1.3, threshold 109.4°F (not charging) / 116.6°F (charging).
- 2 overheat events, both while charging (purple dots, right at the charging-threshold line, peak 117.0°F).
- RSRP/RSRQ around those events sit mostly in Good/Fair — no sustained poor-signal period.

Ref Comparison (Biocore Pro V1) tab — this device (red) vs. a healthy reference unit of the same HW platform (blue), split by charging state:



Both overheating events happened **while charging** (top chart, max 117.0°F vs. the 116.6°F threshold); **zero** overheating events while discharging (bottom chart). Since the reported complaint happened while the device was *worn* — and a device is normally not being charged while worn — this device's root cause is **not yet conclusive**.

Device 1325061716 (RMA 14768) — Biocore Pro V2

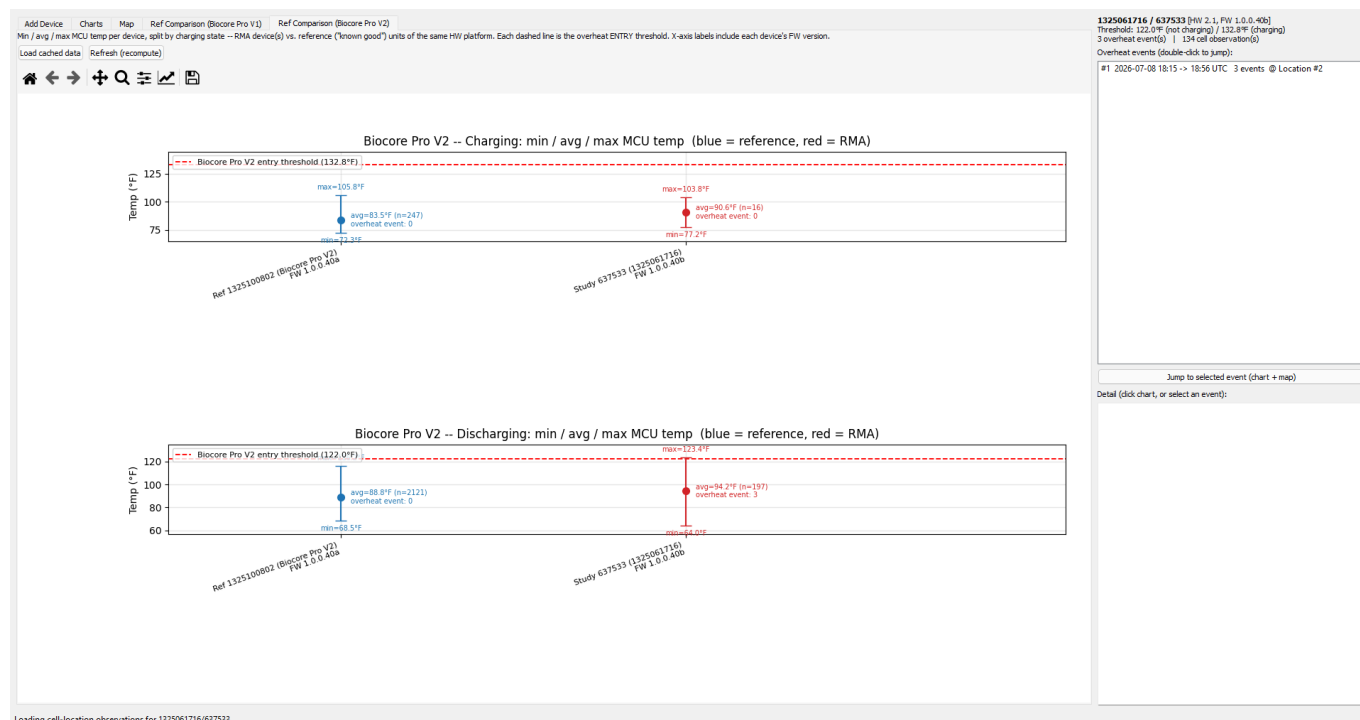
Same flow: Device ID **1325061716** → Look up RMAs → select RMA **14768** → Use this RMA → Study Candidates → Download (candidate **637533** this time had its logs already on disk, so it just skips straight to loading).



- **HW 2.1**, threshold **122.0°F (not charging)** / **132.8°F (charging)**.
- **3 overheating events**, all **while discharging** (red dots, right at the not-charging threshold line).

- RSRP/RSRQ show **sustained poor signal** (gray/red bands) through most of the logged period, not just around the events.

Ref Comparison (Biocore Pro V2):



All 3 overheat events happened **while discharging** (bottom chart, max 123.4°F vs. the 122.0°F threshold); 0 events while charging. Combined with the sustained poor signal on the Charts tab, this points to **weak signal quality as the likely contributing factor**.

Putting it together → drafting the reply

We reviewed the logs for the 3 RMA devices. Device 1324050352 is Biocore Pro V1, the other two (Device 1325061716, Device 1325062089) are Biocore Pro V2. Since V1 and V2 have their own overheat threshold, we compared each group against a reference device of the same Hardware platform.

1. Biocore Pro V1 – Device 1324050352

Device 1324050352 overheated twice while charging (peak 117.0°F, just above the 116.6°F threshold). Signal quality around these events is mostly Good/Fair, no sustained poor-signal period. Also, since these events happened while charging, they are likely not related to the reported complaint — the complaint occurred while the device was worn on the patient, and a device is normally not being charged while worn. So the root cause for this device is not yet conclusive.

2. Biocore Pro V2 – Device 1325061716 and Device 1325062089

Both devices overheated while discharging (1325061716: 3 events, 1325062089: 5 events). Checking each device individually, both show sustained poor signal quality through most of the logged period.

Conclusion

- Device 1324050352: overheated only while charging, not while worn, likely unrelated to the reported complaint. Root cause not yet conclusive.

- Device 1325061716 and 1325062089: overheated while discharging, both with sustained poor signal quality — weak signal is the likely contributing factor.